

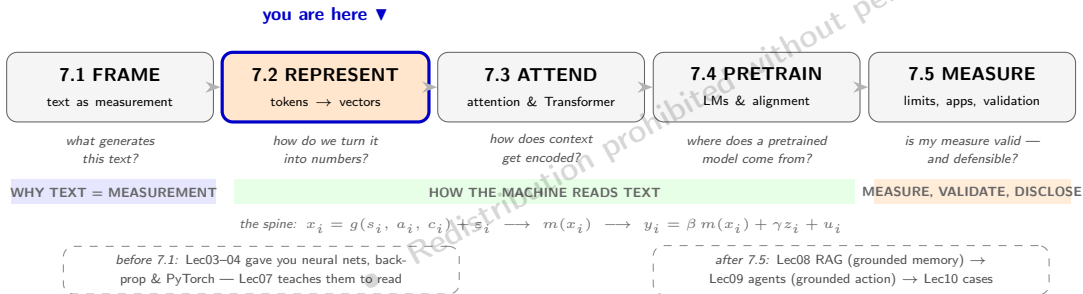
AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.2b: Word Embeddings

● Zhigang Feng

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Lecture 7 in One Map



Route: from counts to coordinates (part 2/2 of REPRESENT) — Word2Vec and GloVe, the geometry of meaning, what a direction in the space can measure, and where static vectors break.

From Counts to Coordinates

Learning a vector space in which meaning has geometry

Why Embed Tokens into Vectors?

- Neural networks operate on real-valued vectors, not discrete symbols.
- An **embedding** is a learned function

$$\mathcal{E}: \{1, \dots, M\} \longrightarrow \mathbb{R}^d,$$

mapping each token ID to a dense vector of dimension d (typically $d = 128$ to 4096).

- **What we want from embeddings:**

- Semantically similar tokens should have nearby vectors
- Algebraic operations (sums, differences) should reflect semantic relations
- The mapping should be *learned* from data, not hand-designed

- **Bonus:** we will also need a *positional* signal so the model knows token order—more on this when we meet the Transformer (Act 3).

Three Approaches to Word Representation

Method	Type	Idea	Year
TF-IDF	Sparse, frequency	Re-weight word counts	1970s
Word2Vec	Dense, predictive	Predict neighbors in window	2013
GloVe	Dense, statistical	Factor co-occurrence matrix	2014
Transformer embeddings	Dense, contextual	Vary by surrounding tokens	2017+

Word2Vec: Dense Embeddings from Context

- **Distributional hypothesis** (Firth, 1957): “*You shall know a word by the company it keeps.*”
- **Idea** (Mikolov et al., 2013): for each word in a corpus, look at a window of $\pm L$ neighboring words. Learn a vector for the center word that *predicts* its neighbors (*Skip-gram*) or learn one that *is predicted by* its neighbors (*CBOW*).
- **Training output:** a single static vector $\mathbf{u}_w \in \mathbb{R}^d$ for each word w in the vocabulary, with d typically 100–300.
- **Geometric magic:** after training, simple vector arithmetic captures analogies, e.g.

$$\mathbf{u}_{\text{king}} - \mathbf{u}_{\text{man}} + \mathbf{u}_{\text{woman}} \approx \mathbf{u}_{\text{queen}}.$$

Words living on the same semantic axis differ by approximately the same vector.

Skip-Gram in One Slide: The Math

- Each word w has two learned vectors: an input vector $\mathbf{u}_w$ and an output vector $\mathbf{v}_w$ (often merged after training).
- For a center word w_t and a context word w_c in its window, the model predicts

$$P(w_c | w_t) = \frac{\exp(\mathbf{u}_{w_t}^\top \mathbf{v}_{w_c})}{\sum_{j=1}^M \exp(\mathbf{u}_{w_t}^\top \mathbf{v}_{w_j})}.$$

- Training objective (negative log-likelihood over all (center, context) pairs):

$$\mathcal{L} = - \sum_t \sum_{c \in \text{window}(t)} \log P(w_c | w_t).$$

- Negative sampling (SGNS)** makes this tractable — instead of the full-vocabulary softmax, push up the dot product of each *true* (center, context) pair and push down $K \in [5, 20]$ randomly drawn pairs:

$$\mathcal{L}_{\text{SGNS}} = \log \sigma(\mathbf{v}_c^\top \mathbf{u}_t) + \sum_{k=1}^K \mathbb{E}_{n_k \sim P_{\text{noise}}} [\log \sigma(-\mathbf{v}_{n_k}^\top \mathbf{u}_t)].$$

“inflation” near “persistent”: pushed together; “inflation” vs. “basketball”: pushed apart.

Skip-Gram: Walkthrough on “central banks target inflation”

- **Sentence:** “central banks target inflation”, window radius $L = 2$, center word = *target*.
- Training pairs formed from this center word:

$(target, central), (target, banks), (target, inflation).$

- For each pair, the score is the dot product $\mathbf{u}_{target}^T \mathbf{v}_{context}$. Gradient descent *increases* this score for observed pairs and *decreases* it for random (negative) pairs.
- **Across the whole corpus:** *target* appears near *rates, inflation, policy, Fed*. So do *raise* and *cut*. Consequently $\mathbf{u}_{target}$ ends up close to $\mathbf{u}_{raise}, \mathbf{u}_{cut}$ and far from $\mathbf{u}_{apple}$.
- **Takeaway:** analogies emerge because consistent co-occurrences place related words at similar positions in $\mathbb{R}^d$.

GloVe: Count Once, Then Fit

Word2Vec slides a window and predicts neighbors; GloVe takes a shortcut — count how often each pair of words co-occurs across the whole corpus, then fit vectors to reproduce those counts.

- Let X_{ij} = the number of times words i and j appear near each other in the entire corpus. (*Fed* and *rates* might co-occur 820 times; *Fed* and *apple*, just 3.)
- Pennington et al. (2014) ask each pair of vectors to reproduce its co-occurrence count:

$$\underbrace{\mathbf{u}_i^\top \mathbf{v}_j}_{\text{how aligned the vectors are}} + \underbrace{b_i + b_j}_{\text{each word's baseline}} \approx \underbrace{\log X_{ij}}_{\text{how often they co-occur}}$$

- The whole model in one sentence:** *two words that keep each other's company should get vectors that point the same way.* Everything else is just fitting that sentence by least squares.

GloVe: Read It Like a Gravity Equation

An economist has met this equation before — it is the gravity model of trade, term for term.

$$\underbrace{\log(\text{trade}_{ij})}_{\text{how much } i, j \text{ trade}} \approx \text{FE}_i + \text{FE}_j + (\text{pair closeness}) \iff \log X_{ij} \approx b_i + b_j + \mathbf{u}_i^\top \mathbf{v}_j$$

- $\log X_{ij}$ — how much the two words “trade” with each other. We take logs because co-occurrence counts span orders of magnitude (3 vs. 820) — the same reason we use $\log$ GDP.
- b_i, b_j — **word fixed effects**. *the* co-occurs with everything simply because it is everywhere; the fixed effects soak up that baseline popularity so we don’t mistake it for meaning.
- $\mathbf{u}_i^\top \mathbf{v}_j$ — what is *left over* once popularity is removed: the pair’s **specific affinity**. This, and only this, becomes meaning.

The payoff: words that co-occur more than their popularity alone would predict get vectors pointing the same way; words that never meet stay at right angles. *Meaning becomes geometry, estimated like a regression.*

GloVe at Work: A Central-Bank Corpus

Watch “monetary-policy vocabulary” fall out of nothing but co-occurrence counts.

co-occurs with →	<i>rates</i>	<i>inflation</i>	<i>apple</i>
Fed	820	410	3
ECB	540	390	2
hawkish	210	160	0

What the fit does, row by row:

- *Fed* & *rates* co-occur 820 times ($\log X \approx 6.7$). The baselines $b_{\text{Fed}} + b_{\text{rates}}$ cannot explain a number that big — the only way to fit it is to make $\mathbf{u}_{\text{Fed}}$ and $\mathbf{v}_{\text{rates}}$ **point the same way**.
- *Fed* & *apple* co-occur just 3 times ($\log X \approx 1.1$) — about what the baselines predict for *any* two words. Nothing is left to explain, so their vectors end up **at right angles** (unrelated).
- *Fed*, *ECB*, and *hawkish* all keep the *same company* (rates and inflation, never apple), so all three get pulled into the **same corner** of the space — to the geometry they are near-synonyms.

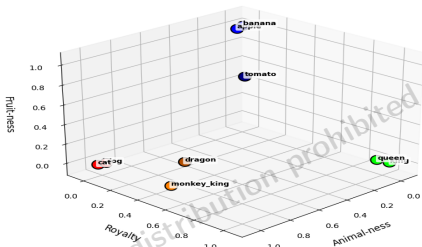
The punchline: nobody told the model what a central bank is. Rows with similar co-occurrence patterns are forced to similar coordinates, and a “monetary-policy” cluster emerges purely from counting.

The Geometry of Meaning

Distances, directions, analogies — and the biases baked in

A 3D Toy of Word Embeddings

Enhanced 3D Word Embeddings Visualization



The three axes loosely correspond to “Animal-ness”, “Royalty”, and “Fruit-ness”. Words like *cat*, *king*, and *apple* cluster along the dimension that fits them best. Real LLM embeddings live in hundreds of dimensions, but the same intuition holds.

We compare words by the *direction* of their vectors — **cosine similarity** $\cos(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ — not raw distance: frequent words grow longer vectors in training, and direction strips that out.

Analogies and the Bias Problem

- **Geometric magic** (Mikolov et al., 2013): simple vector arithmetic captures analogies.

$$\mathbf{u}_{\text{king}} - \mathbf{u}_{\text{man}} + \mathbf{u}_{\text{woman}} \approx \mathbf{u}_{\text{queen}}$$

$$\mathbf{u}_{\text{Paris}} - \mathbf{u}_{\text{France}} + \mathbf{u}_{\text{Italy}} \approx \mathbf{u}_{\text{Rome}}$$

- Words living on the same semantic axis differ by approximately the same vector — gender, capital city, comparative tense, monetary stance. (Caveat: the clean linear pattern weakens for specialized, low-frequency vocabulary — validate on your own terms, don't extrapolate from benchmark word lists.)
- **Bias is encoded too** (Bolukbasi et al., 2016, *NeurIPS*; Caliskan, Bryson & Narayanan, 2017, *Science*):
 - $\mathbf{u}_{\text{man}} - \mathbf{u}_{\text{woman}}$ has a large component on *programmer* vs. *homemaker*.
 - Word embeddings reproduce IAT-style biases at human magnitudes.
 - For economic text: occupation/gender, race/credit-risk, region/sentiment correlations all transfer from the training corpus into the geometry.
- **Lesson:** embeddings are a mirror of the corpus. Validating that a sentiment vector reflects *policy* stance and not *speaker identity* is a real research step (returns in T5b). Flip side: the encoded

Four Generations of Representation, One Table

Every step toward “understanding” trades away interpretability and economy.

	Counts	TF-IDF	Word2Vec	Contextual (BERT)
Dimension	$V \sim 10^4\text{--}10^5$	same	$d = 100\text{--}300$	$d = 768\text{--}1024$
Sparsity	>99% zeros	>99% zeros	dense	dense
Semantic similarity	none (orthogonal)	weak (IDF re-weight)	yes (distributional)	strong (context-aware)
Polysemy	no	no	one static vector	varies with context
Interpretability	high (word = axis)	high	low (latent space)	very low
Data needed	none	none	$> 10^8$ tokens	$> 10^{10}$ tokens
Compute	trivial	trivial	moderate	high
Typical econ use	EPU, dictionaries	retrieval, similarity	product networks, culture scores	sentiment, stance, extraction

Rule: when external validity rests on an auditable, line-by-line evidence chain, the fancier representation is often the *worse* choice. Pick the rung your research design needs — not the newest one.

From Analogies to Constructs: The Hawk–Dove Direction (Thread B)

Define an economic concept as a direction in the space.

- Candidate monetary-stance axis: $\hat{\mathbf{v}} = \mathbf{u}_{\text{hike}} - \mathbf{u}_{\text{cut}}$. Score any FOMC passage by the cosine of its (averaged) vector with $\hat{\mathbf{v}}$.
- The same trick, generalized: culture scores from seed-word directions (Li, Mai, Shen & Yan 2021); time-varying product-market networks from 10-K business-description similarity (Hoberg & Phillips 2016).
- **Validation idea:** a genuine stance direction should co-move with policy — check the $\hat{\mathbf{v}}$ -scores against subsequent fed-funds-rate changes before using them in a regression.

The separability assumption. *king* – *man* + *woman* works because the difference is (mostly) one clean dimension — everything else cancels. But *hike* co-occurs with overheating and *cut* with recessions, so $\hat{\mathbf{v}}$ may encode **stance and the business cycle at once** — like a bond that bundles duration and credit risk in one price. Purge or control for the cycle dimension before calling it “policy stance”; ask which institutional features make the coupling worse.